Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

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Abstract

We address catastrophic forgetting in continual key-information extraction from documents: replaying a few hundred stored documents comes close to joint training on average when the extractor meets new domains, yet fails when its label space grows. We study this failure as an investigation rather than a method. On a class-incremental stream of receipts, we test a ladder of hypotheses, state the prediction of each in advance, and measure where the errors go. First, forgetting is not simply harder: with a whole-document memory and a shared softmax, the model barely learns the last sessions it is trained on, which naive training learns well. Then, stale stored labels suffice to explain nearly all of it: giving every stored document its true labels before each session, an oracle no continual learner has, brings the same shared softmax to within 2.4 *points of joint training on every seed (*92.1 *against* 92.0 *full-label F1), although the current documents still call old entities "O". Without true labels, an unbiased loss in the style of MiB recovers about half of the full-label gap, whereas stopping the sharing of the background class closes most of it, as existing per-session readouts do: with the same memory, one background logit per session reaches* 89.1 *full-label F1 (*88.4 *with raw documents in its place). We argue that an "O" label is evidence only against the classes its annotator could see, a principle familiar from training on partially annotated data. Finally, where label growth supplies the routing key, on one schema, per-session low-rank trunk updates can replace the memory; across schemas they did not, in one run.*

1. Introduction

Key-information extraction (KIE) labels each OCR token of a visually-rich document (form, invoice, receipt) using its words, its 2-D layout and the page image, and multimodal encoders such as LayoutLMv3 [13], LiLT [37] and BROS [11] define the state of the art. However, deployed extractors meet new document types and growing schemas, while retraining on all history is blocked by cost and by privacy constraints. This is a continual-learning (CL) problem, whose central obstacle is catastrophic forgetting [24].

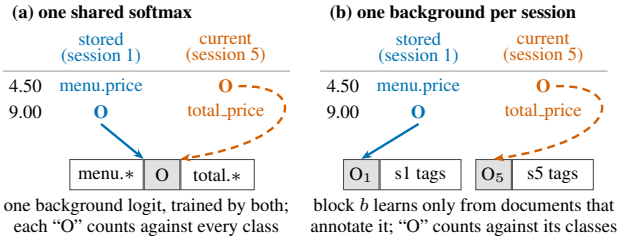


Figure 1. **The background conflict on one pair of price tokens.** A receipt stored in session 1 labels its total "O"; a current receipt of session 5 labels its item price "O". (a) With one shared softmax both "O" labels train the same background logit, against every class. (b) With one background per session (blocks 1 and 5 shown, each with its session's tags), an "O" is evidence only against the classes its annotator could see.

There exist many CL methods to cope with forgetting, and replay is empirically the strongest [23, 38]. On documents it looks sufficient: on LayoutLMv3, plain replay (ER) with 200 stored documents learns a stream of forms and receipts under one fixed label space to 87.9 AA, within 3.5 points of joint training on every seed (mean 88.7; Appendix E). However, when the label space grows, by adding CORD's 30 receipt fields six at a time, the same method (200 documents) reaches 24.3 AA, against 91.3; both ER cells are legacy-protocol runs (†) that early-stop on the test split, which favours ER (Sec. 3). Distillation, prompts, low-rank adapters and the continual-NER methods built for the shift of the "O" label do no better (Tab. 3). A class-incremental stream is expected to be harder; what is not obvious is why a memory holding the old documents does not help.

In this work, we treat this question as an investigation (Tab. 1): for each explanation a reader would offer, we state a prediction before its test, run an intervention on the same encoder, stream and recipe, and measure where the final model's errors go (Fig. 2). First, forgetting is not just harder (H0): with a whole-document memory that retains under domain shift, the model suppresses the last sessions it is trained on, which naive training does not do. Then, stale labels in the memory (H1) suffice to explain nearly all of the failure: relabelling every stored document before each session, an oracle, brings the shared softmax to within 2.4

points of joint training on every seed, although the current documents still call old entities "O" (H2). Without true labels, a MiB-style unbiased loss recovers about half of the full-label gap, whereas removing the shared background (H3, Fig. 1) closes most of it, given a memory. Finally, per-session isolation can replace the memory (H4), but only where label growth supplies a routing key.

None of the ingredients is new. The background shift is known from class-incremental segmentation [4, 6, 7], incremental detection [21, 27] and continual NER [22, 28, 43], and separated or per-session softmaxes [1, 14, 20] already undo the shared normalisation. However, no prior work takes apart, on token-labelled documents, which part of the failure a memory repairs and which survives true stored labels. The contributions of our study are as follows: (1) we decompose why replay fails under label growth with a memory $\times$ readout factorial, relabelled memories that isolate the stored-label side, and where the errors go under each readout (Sec. 4); (2) we report an empirical finding, not a new mechanism: a per-session background readout, as in ICE-O and MORE, combined with whole-document replay and a supervision rule in the spirit of training on disjoint tag sets, closes most of the gap to joint training in continual KIE on every seed, where the continual-NER and replay methods in our comparison do not; it works with a raw or a latent memory, and the latent one is within 1.7 full-label points of raw documents on every seed while keeping no pixels or token ids (Secs. 5 and 6); (3) we delimit when the memory can go: with the conflict removed, per-session low-rank trunk deltas (SBR-Recall) replace the memory on a single-schema stream, within 2.1 points of joint training on every seed under both metrics, and fail without a memory across annotation schemas (one seed), where the task-identity view says they must (Sec. 5.3); on images, which have no background class, the latent memory is competitive with raw replay while SLCA and RanPAC lead (Appendix F).

2. Related Work

Related studies fall into three groups: background shift, separated readouts and continual document IE.

Background shift. When each step annotates only its own classes, the background label changes meaning. In class-incremental segmentation MiB [4] lets a background label mean background or any old class in the loss, and the old model's background mean background or any new class in the distillation; PLOP [7] and SSUL [6] pseudo-label the background with the old model. Incremental detection has named both its stored side [21] and its two-sided form [27]. In continual NER the shift of "O" is the unlabelled-entity problem [22], answered by distillation on "O" tokens [28] and pseudo-labels [43]. We do not claim the conflict; instead, we separate its stored and current sides with relabelled memories and an unbiased loss and measure what each leaves.

Separated readouts and task identity. A softmax over old and new classes together biases them, and separating it removes the bias [1]; ER-ACE [3] masks the old classes out of the softmax of incoming samples. A "none of mine" output per task turns class-incremental learning into within-task prediction plus task identification [15]. For instance, MORE [14] isolates each task's parameters and trains each head's out-of-distribution output from saved samples; as another example, ICE-O [20], on a frozen encoder with no memory, trains one softmax per session over a constant Other logit. SBR is this pattern with a learned background per session and old blocks kept trainable from a memory; SBR-Recall shares MORE's isolation, by low-rank trunk deltas, with no saved samples.

Continual document IE. Document-level continual learning is recent and small: multimodal LLMs forget under fine-tuning [42], GDI-Bench [18] freezes most parameters, continual table detection [26] uses plain ER, ProtoNER [17] uses prototypes and AMD-Proj [33] projects gradients against task subspaces. Recently, Sassioui et al. [34] benchmarked layout-aware models by stream regime and found that no method they test consistently preserves old knowledge under label-space shift. However, they decompose forgetting by regime, whereas we decompose it by cause. Our memory is a form of latent replay [9, 29] that keeps whole documents; the broader CL families are surveyed by Wang et al. [38].

3. Setup

Streams and models. A document is its OCR tokens, normalised boxes and page image; the model emits BIO tags per token, scored by entity-level F1 (`seqeval`). We consider three streams (Tab. 2; sessions in Appendix A). **CIL-CORD**, the stream of every rung: CORD's 30 receipt classes arrive six at a time over five sessions; the head grows, and every entity outside the session, earlier or later, is labelled "O". **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema. **Mixed**: six sessions over three datasets and annotation schemas. LayoutLMv3-base [13] is the encoder of every rung; LiLT [37], BROS [11] and text-only BERT appear on DIL.

Memory and comparison set. Unless stated, a memory is CoLaR (Sec. 5.2): 50 documents per session stored as rank-128 SVDs of their layer-4 activations. We benchmark against two references (naive fine-tuning, joint training) and eight methods from five families (Tab. 3): ER [32], DER++ [2], LwF [19], CODA-Prompt [35], CL-LoRA [10], LCA [36],

Table 1. **The hypothesis ladder** (CIL-CORD, LayoutLMv3 unless stated; mean ± s.d. over three seeds, per-seed values when fewer; $^{n=k}$: k seeds so far; †legacy protocol; "–": no full-label run). Verdict: what the row says about its rung's prediction, on every seed; "open": only preliminary, cross-protocol evidence so far. ^{r}True labels for the documents stored before the session just finished, at each boundary, or for every stored document before each session (Sec. 4.2). ^{e}ER also trains every layer and keeps 200 raw documents, so it changes more than the memory format (Sec. 4.1). ^{w}Trains every layer, where the CoLaR rows freeze the embeddings and layers 1..4. The pre-session row of H1 also tests H2: it leaves every current document stale.

Rung	Hypothesis	Prediction if true	Intervention (CIL-CORD unless stated)	Full-label F1	AA	Verdict
			Naive fine-tuning (reference)	13.0±0.2	18.3±0.5	
H0	Only drift: forgetting is just harder	a memory that retains under domain shift retains here; raw beats compressed	whole-document memory (CoLaR), shared softmax	30.7±1.5	33.5±1.0	refutes
			raw-document memory (ER), shared softmax^{e}	–	24.3±2.4†	refutes
			raw-document memory, blocked readout^{w}	88.4±1.0	88.4±1.1	does not support
H1	Stale labels in the memory	true labels in the stored documents close the gap	older stored documents relabelled at each boundary^{r}	76.5±2.3	63.6±1.9	partly
			every stored document relabelled before each session^{r}	92.1±1.3	91.3±1.3	supports
H2	Current documents mislabel old entities	a loss that lets "O" mean "O or an old class" closes the gap	replay and current losses sliced to their classes	–	60.5±10.1†	open
			MiB-style unbiased CE, both directions	60.0±1.2	72.1±1.9	partly
H3	The shared background normalisation	one background per session closes the gap without true labels, given a memory	CoLaR memory, session-blocked readout (CoLaR-SBR)	89.1±2.0	88.4±2.6	supports
			session-blocked readout, no memory	11.4±2.3	21.6±2.2	supports
			CoLaR memory, blocked readout with a constant background logit	82.4±6.9	80.2±8.1	partly
H4	Isolation replaces memory where a routing key exists	per-session deltas need no memory on one schema, and fail across schemas	per-session deltas, blocked readout, no memory (SBR-Recall)	92.6±1.0	91.9±0.6	supports
			ICE-O (our port): frozen encoder, head per session, no memory	57.0±0.7	35.2±0.7	partly
			the same on Mixed (AA), no memory	–	44.7$^{\dagger,n=1}$	supports
			the same on Mixed (AA), 50-document memory	–	85.9$^{\dagger,n=1}$	supports
			Joint training (reference)	92.0±0.4	91.3±0.8	

and the continual-NER methods CPFD [43] and IS3 [30]; all are our ports (Appendix C).

Protocol. We train with AdamW, learning rate 5×10^{-5} (LCA keeps its own SGD recipe; Appendix C), batch size 2 and seeds 42, 7 and 123. Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100) and early-stops on 10% held out from its *training* data; the test split never selects a model. Cells marked † come from earlier runs (*legacy protocol*) that early-stop on the test split and vary in epoch cap and precision. A cell is clean only when all three clean seeds exist (Appendix A).

Scoring. Our headline number is the *full-label F1*: the final model on the complete CORD test set with all 30 classes, as in continual NER [22]. Only clean runs record it; against

Table 2. **Benchmark settings** (sessions: Appendix A). Train/test documents: FUNSD 149/50, SROIE 626/347, CORD 800/100. AA: mean F1 of the final model over the sessions; full-label F1: on the unmasked CORD test set (Sec. 3).

	CIL-CORD	DIL	Mixed
Datasets	CORD	FUNSD, SROIE, CORD in turn	FUNSD, SROIE, CORD
Sessions	5	3	6
Label space	grows, 6 per session	fixed, unified	grows, 3 schemas
Classes (BIO tags)	30 (61)	4 (9)	12 (25)
At a boundary	new classes	new domain	new classes, domain or schema
Reported	full-label F1, AA	AA	AA

a cell with only AA (†, DIL, Mixed) we compare in AA. Retention needs per-session test splits, whose labels show only that session's classes; from $R[i, j]$, the F1 on session j after session i, AA $= \frac{1}{T}\sum_j R[T, j]$. Since raw scoring of such labels charges a model for every correct entity of another session, we score each session *session-restricted*, mapping a predicted tag outside session j's label set to "O" before span matching. On every seed both metrics order the main groups of clean runs the same way (Appendix A). Finally, tuning and evaluating on the same scenario overstates CL methods [5]; hence the comparison set was fixed after a first grid on all three streams, the memory settings were chosen on DIL, SBR's claim threshold (0.5) was not tuned, and the one setting chosen on CIL-CORD is SBR-Recall's delta capacity (Appendices C and D).

4. The investigation

Every rung runs on CIL-CORD with LayoutLMv3 under the clean protocol and changes one thing against the run it is compared with, unless stated (H0's memory rows change several). Table 1 lists the rungs and Fig. 2 draws them; a verdict word is one the numbers justify on every seed.

4.1. H0: only harder forgetting

Prediction. We first hypothesize that label growth only makes drift harder. Then the memory that retains under domain shift would still retain here, a raw memory would do better, and the losses would look like naive forgetting, where old sessions are lost and the newest is learned well.

Test and result: refuted. The CoLaR memory comes within 2.2 AA points of joint training on DIL on every seed (means $87.8^{\dagger}$ against $88.7^{\dagger}$). However, on CIL-CORD with a shared softmax it reaches 30.7 ± 1.5 full-label F1 (33.5 ± 1.0 AA) against 92.0 ± 0.4 (91.3 ± 0.8). Raw documents do no better (ER, 24.3 $\text{AA}^{\dagger}$, below every CoLaR seed), but ER also trains every layer, keeps 200 documents and is a legacy run; the raw-memory comparison under the same readout is in Sec. 5.2. What separates the memory run from naive forgetting is the session being trained: its newest session ends at 2.0 to 16.0 F1, whereas naive training learns every session to at least 78.6 on every seed. It is noted that the memory run also freezes layers 1..4 after the first session and replays four stored documents per step; with the same freezing (and, with the memory, the same replay), the blocked readouts with and without the memory learn every session to at least 81.1 (per-seed minimum), so lost plasticity does not explain the suppression. Besides, it keeps 18.7 to 22.6% of the old sessions' spans where naive training keeps none, at an old-span precision of at least 71.6% (a lower bound; Appendix A).

What it rules out, and what it leaves. Hence, drift alone does not explain the failure: the memory that answers drift on DIL leaves the session being trained suppressed here, which naive training, with all of its drift, does not show. Nevertheless, drift is real: naive training keeps nothing of the old sessions, on a fixed label space every encoder forgets its earlier domains and replay, not EWC or LwF, repairs it; and under naive fine-tuning the drift is largest near the output on all four encoders (Appendix E).

4.2. H1: stale labels in the memory

Prediction. We next hypothesize that the stored labels are stale. A document stored at session t carries "O" on every entity of another session, earlier or later, so the memory itself calls background what the model must keep or learn; in detection, the same staleness of replayed examples was named foreground shift and blamed for replay's failure [21]. If these stale labels are the cause, giving the stored documents their true labels should close the gap with the shared softmax unchanged.

Test. We refresh the stored labels at two points. First, *at each boundary*, every document stored *before* the session just finished receives the true labels of all classes seen so far; that session's own documents keep their labels until the next boundary, and no such refresh covers the session trained next. Then, *before each session*, every stored document receives the true labels of all classes in the head grown for that session, so no stored label is stale while a session trains; the current documents still call old entities "O". Both read labels no continual learner has and are diagnostics, not methods.

Result: partly at the boundary, supported before each session. At the boundary, against the same memory without the refresh, the refresh lifts full-label F1 from 30.7 to 76.5 and AA from 33.5 to 63.6 (means, three seeds): 72 to 79% and 48 to 56% of the seed-matched gap to joint training (92.0, 91.3), and still below CoLaR-SBR under both metrics. What it cannot reach is the session being trained, whose classes are "O" in every stored document: sessions 3 to 5 are

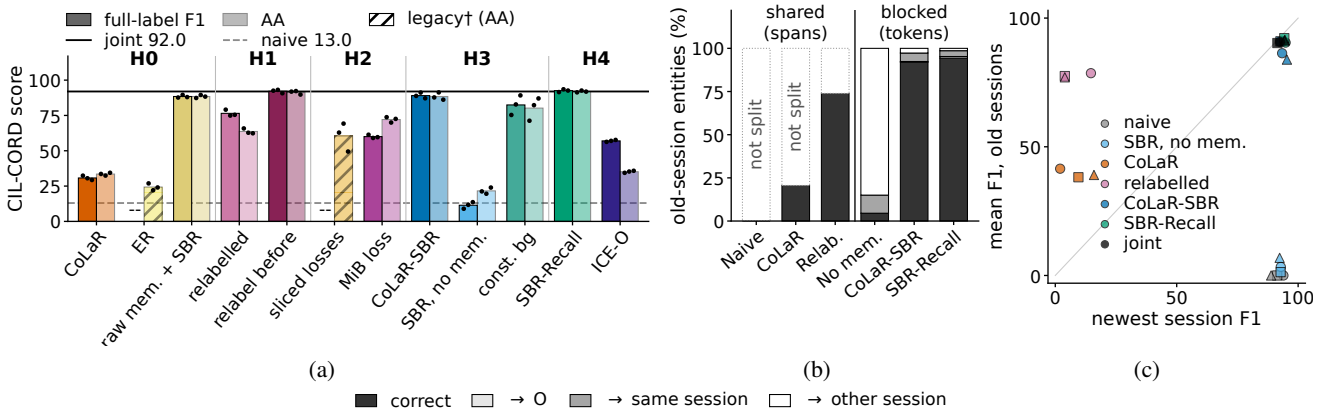


Figure 2. **The findings in one figure** (CIL-CORD, LayoutLMv3, clean protocol unless marked). (a) The ladder (Tab. 1): full-label F1 (dark) and AA (light); bar: mean of three seeds, dots: seeds; hatched: legacy (†), AA only; "–": no full-label run; lines: joint and naive full-label F1. (b) Where the final model puts old-session gold entities (sessions 1 to 4, mean over seeds); a shared softmax records only span recall. (c) Per seed (circle: 42, square: 7, triangle: 123), newest-session F1 against mean old-session F1: with a shared softmax and a memory the newest session is suppressed; naive training learns it and keeps nothing older. More pictures: Appendix D.

learned only to 2 to 37 F1 on their own diagonal, sessions 3 and 4 rise to 74 to 88 one session later, and the last ends at 3.9 to 14.7, a fifth of AA but 8% of the test spans. In contrast, refreshed before each session, the same memory with the same shared softmax reaches 92.1±1.3 full-label F1 and 91.3±1.3 AA, within 2.4 points of joint training on every seed under both metrics and above CoLaR-SBR on every seed (Tab. 1). As a result, stale stored labels suffice to explain nearly all of the failure of replay under a shared softmax, on the session being trained as much as on old ones (three seeds). However, the control is an oracle: it shows what the memory must hold, not how a continual learner could obtain it.

4.3. H2: current documents mislabel old entities

Prediction. We then hypothesize that the current documents are the cause: each labels every entity of an earlier session "O", so even a memory with true labels is opposed in every batch by documents that call the same strings background; this is the second side of the conflict named in detection [27]. If this pull is the cause, a loss that lets a current "O" mean "O or an earlier class", as MiB does for segmentation [4] and marginal-likelihood taggers do offline [8], should close the gap within a shared softmax.

Test. We use two clean controls: the never-stale memory of H1, which leaves every current document stale, and, with the memory, the shared softmax and no true labels, MiB's unbiased cross-entropy in both directions: a current "O" token matches "O" or any earlier-session class, a replayed one "O" or any class introduced after storage (no distillation).

Result: not needed, and a loss repair only partly helps. With true stored labels the gap closes although every current document still calls old entities "O" (H1); hence the current side is not needed to explain the failure. The MiB-style loss lifts full-label F1 to 60.0±1.2 and AA to 72.1±1.9, above CoLaR and below joint training on every seed under both metrics: 46*to*49% of the seed-matched gap in full-label F1 and 64*to*70% in AA, and below CoLaR-SBR on every seed. It learns every session to at least 90.1 on its diagonal, removing the suppression of the session being trained, but the first session ends at no more than 35.5 F1 on any seed.

4.4. H3: the shared normalisation

Prediction. Finally, we hypothesize that the conflict lives in the shared normalisation, one background logit trained by documents that disagree about what is background. Then one background logit per session, normalised with that session's tags only, should close most of the gap with no true labels. This is SS-IL's diagnosis of a shared softmax [1] with a background added, and ICE-O's readout [20] without its frozen encoder. It needs the memory, since separating the labels does not undo drift; thus without a memory the same readout should fail, and fail the way drift does.

Result: supported, on every seed. With the memory, the session-blocked readout (CoLaR-SBR, Sec. 5.1) reaches 89.1±2.0 full-label F1 and 88.4±2.6 AA, 0.6 to 5.3 full-label points below joint training per seed, above the memory relabelled at each boundary and below the memory relabelled before each session on every seed, under both metrics. The owning block claims 95.4 to 98.3% of the old entity tokens, and most remaining errors (56.6 to 71.6% of the wrong tokens) give the token another tag of its own session. On

the other hand, without the memory the same readout scores 11.4±2.3 (21.6±2.2 AA), and its errors go elsewhere: not one old entity token is predicted "O", 83.8 to 93.5% of the wrong ones receive another session's class, and each old block claims 17.0 to 24.3% of the tokens of its own split that are not its entities, against 0.6 to 1.7% with the memory. Old blocks on a drifted trunk claim far more than their own entities; that is drift, and the memory reduces it.

What goes against it. It is neither the only repair nor the best one: true stored labels before each session do better (Sec. 4.2), but the blocked readout needs none. The readout does nothing alone, and it is not free: on DIL, with nothing to separate, CoLaR-SBR is below CoLaR (AA 86.2† against 87.8†).

What discriminates what. Overall, drift (H0) is told apart from the label hypotheses by the session being trained, suppressed under a shared softmax with a memory and learned without one. We read the shared normalisation (H3) as the channel through which stale labels act, as negatives for every class; then separating it should work even with a background that does not learn. With every background logit fixed at zero, as in ICE-O, the memory and blocked readout stay above CoLaR and below joint training on every seed under both metrics, recover most but not all of CoLaR-SBR's gain over CoLaR on seeds 42 and 7, and are above CoLaR-SBR on seed 123 (Tab. 1). Hence the reading is partly supported: separation does most of the work, and the learned background adds to it on two of three seeds.

5. From diagnosis to design

The principle. The rungs point at one fact about annotation under label growth, familiar from training on partially annotated data [8]. A token marked "O" in session s means "none of the classes annotated in session s": it is evidence against those classes and silent about every other. However, a shared softmax turns it into evidence against all classes at once (H3), and stale stored labels are the same error made by the memory (H1). Hence we follow one principle: *supervise each class only with negatives from documents that annotated it, and normalise together only classes that were annotated together.* Three consequences follow: each session needs its own background; the memory must keep whole documents with their own labels and the session that wrote them; and each session's background logit, trained on that session's true negatives, is also a detector of its entities, so it can route a token without task identity.

5.1. SBR: one background per session

SBR is our instantiation of a per-session background readout [14, 20] (Fig. 3b). Every session that brings new tags

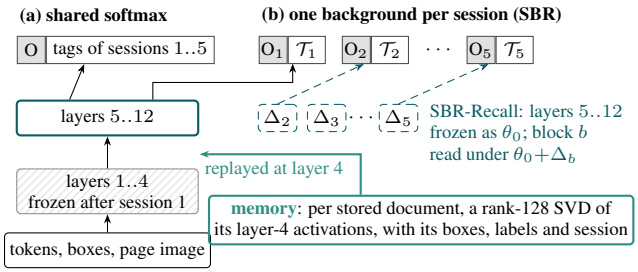


Figure 3. **The two readouts on one encoder and memory.** (a) One softmax over all tags and one background. (b) SBR: one background logit per session, trained only by documents that annotate it (Eq. (2)); the largest claim $e_b = 1 - p_b(\mathrm{O}_b) > 0.5$ labels the token. SBR-Recall drops the memory and reads block b under its session's delta Δ_b (Eq. (3)).

opens a block b holding those tags $\mathcal{T}_b$. Whereas a shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$, SBR gives every session its own background logit $o_b(x) = w_b^\top h(x) + c_b$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\right), \tag{1}$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block b a token labelled "O" in session-b data is truly background, so the block's supervision is never stale. For a token x with label y in a document of session s,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\ \mathrm{CE}\big(p_b(\cdot \mid x),\, y_b\big),$$
$$y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: $m_{sb}{=}1$ when session s annotates every tag of $\mathcal{T}_b$; $m_{sb}{=}1$ with $y_b{=}\mathrm{O}_b$ when b belongs to a different annotation schema (every token is a negative for b, a training-time convention; Appendix B); and $m_{sb}{=}0$ otherwise, because the document's "O" labels say nothing about block b. Stored documents follow the rule of the session that stored them. Since a session that adds no tags creates no block, SBR has a single block on a fixed label space. At inference each block reports its claim $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest e_b labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise, as in ICE-O's "Other unless some session scores above it". Blocks trained at different times need not produce claims on one scale; hence we measure the routing directly (Fig. 2b).

5.2. The memory

SBR's supervision needs stored documents that keep each token's label at its position and the session that wrote it. CoLaR (Compressed Latent Replay) keeps whole documents:

after the first session layers $1..k$ are frozen, and each stored document's layer-k activations H_j are kept as a rank-r truncated SVD with its boxes, mask, labels and session, rebuilt at replay and injected at layer k; it holds no raw pixels or token ids (no privacy claim: activations can leak content). The memory format did not change the finding here (three seeds, CIL-CORD): raw documents under the same readout, with the whole network trainable where CoLaR freezes the embeddings and layers $1..4$, reach 88.4 ± 1.0 full-label F1 against 89.1 ± 2.0, seed-matched differences (raw minus latent) of -1.7 to $+0.9$ points (-1.9 to $+2.2$ in AA). We use CoLaR because it keeps no pixels or token ids (Sec. 6).

5.3. SBR-Recall: the test of H4

Prediction (H4). We hypothesize that, with the conflict removed, the memory is needed only against drift. The task-identity view [15] says when it can go: if each block is read under the trunk state that learned it, nothing drifts for that block, provided the model can tell which state a token belongs to. On one schema under label growth each block's background logit supplies that key. In contrast, across schemas the all-background terms of Eq. (2) train every old block towards background on other-schema documents with nothing to pull it back, so isolation should fail without stored documents; on a fixed label space there is one block and nothing to route by (not run).

The intervention. To test it, SBR-Recall freezes the layers above k after the first session (the base θ_0), stores for each later session b a rank-64 delta Δ_b [12] of them ($\Delta_1{=}0$), and computes every term of Eq. (2) for block b, at training and test time, under the trunk state that learned it (Appendix B):

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b,\, z_{\mathcal{T}_b}\,]\right)\left(x;\, \theta_0 + \Delta_b\right). \qquad (3)$$

Result: supported, with its predicted failure. On CIL-CORD SBR-Recall reaches 92.6 ± 1.0 full-label F1 (91.9 ± 0.6 AA) with no stored document, within 2.1 points of joint training on every seed under both metrics; its owning block claims 97.0 to 98.3% of the old entity tokens. Our port of ICE-O (frozen encoder, one head per session over a constant Other logit, no memory) forgets nothing (final row within 0.3 points of its diagonal on every seed) and reaches 57.0 ± 0.7 full-label F1 (35.2 ± 0.7 AA), below SBR-Recall on every seed under both metrics; under our 10-epoch cap sessions 4 and 5 were still improving and session 3 early-stopped at 0.0 F1 on every seed, so this is a lower bound for ICE-O. On Mixed without a memory, SBR-Recall scores 44.7 AA† (seed 42): as predicted, the blocks of the other schemas fall silent until their domain returns, and with the 50-document memory restored it scores 85.9 (Appendix D).

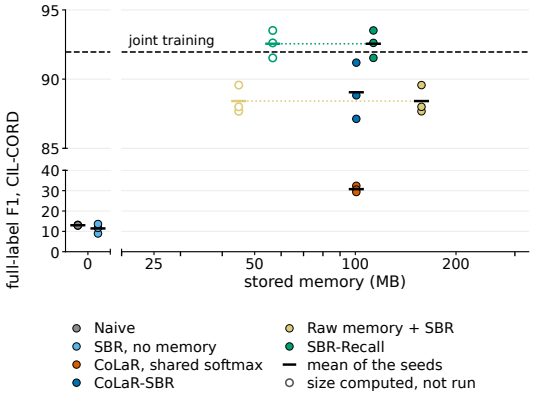


Figure 4. **Accuracy against stored bytes** (CIL-CORD, clean protocol; dots: seeds, bar: their mean). Bytes: what a method keeps across sessions; joint training keeps every document. Filled: the size the run held; hollow: the same run's size computed for an encoding it never ran in (raw documents with 8-bit pixels, SBR-Recall's deltas at two bytes per parameter); a dotted line joins the two sizes of one run. Sizes: Tab. 5.

6. Ablations and memory cost

Which component matters. On the runs of Tab. 1 (gathered in Tab. 7) and on every seed, the memory alone raises full-label F1 over naive training by 16.3 to 19.2 points, and the blocked readout alone moves it by -4.3 to $+0.7$; together they raise it by 74.1 to 78.4, more than the two alone add up to, so the gain is an interaction, not a sum. With raw documents in place of the latent memory (whole network trainable) the finding did not change: raw is neither above nor below CoLaR-SBR on every seed. Per-session deltas in place of the memory are above CoLaR-SBR under both metrics. True labels in the memory, without the blocked readout, stay below CoLaR-SBR when refreshed at the boundaries and are above it when refreshed before each session (Sec. 4.2). Design-choice ablations: Appendix D.

What the memory costs. CoLaR stores 402.7 KB per document (100.7 MB for the 250 documents of CIL-CORD; Fig. 4), $2.8\times$ less than the fp16 layer-4 activations it compresses. Raw replay keeps 630.8 KB per document as our code holds it, with fp32 pixels and int64 ids and boxes (157.7 MB), and would take 44.8 MB with 8-bit pixels. SBR-Recall keeps one delta per session instead: 28.3 MB in fp32 as trained (113.2 MB in all), or 14.2 MB (56.6 MB) at two bytes per parameter, a reporting choice that no run evaluated.

Bytes do not separate the methods; the readout does: every store read through the learned-background blocked readout (latent or raw documents, or deltas) ends within 5.3 full-label points of joint training on every seed, and the same latent bytes under a shared softmax stay at 30.7 ± 1.5. At rank 128 CoLaR saves no bytes: it is $2.2\times$ larger than raw

Table 3. **Benchmark results on LayoutLMv3**, grouped by family. Memory: what a method keeps (sizes on CIL-CORD; SBR-Recall at two bytes per parameter, half its fp32 size). Readout: one shared softmax or one block per session. Mean ± s.d. over three seeds, per-seed values when fewer; "–": not run. †legacy protocol; $^{n=k}$: k clean seeds so far; ^{c}one fp16 LwF seed that collapsed to all-background is excluded; ^{b}with the 50-document memory; ^{p}made before we fixed our port's orthogonality penalty. Ports and protocols: Appendix C.

				CIL-CORD		Mixed	DIL
Family	**Method**	**Memory (what is stored)**	**Readout**	full-label F1	AA	AA	AA
Reference	Naive fine-tuning	none	shared	13.0±0.2	18.3±0.5	44.4/43.7†	41.3±1.5†
	Joint training	all data, all sessions	shared	92.0±0.4	91.3±0.8	86.9/88.5†	88.7±1.1†
Regularisation	LwF (2016)	previous model	shared	–	18.4/18.5^{†,c}	47.7^{†,c}	43.7±1.1†
Replay	ER (2019)	200 raw documents	shared	–	24.3±2.4†	75.6/73.3†	87.9±1.4†
	DER++ (2020)	200 raw documents + logits	shared	–	20.4±1.4†	75.1/74.0†	88.0±0.9†
Prompt, LoRA	CODA-Prompt (2023)	none (prompt pool)	shared	–	14.0/15.9†	12.1/10.6†	32.5±1.2†
	CL-LoRA (2025)	previous model, past adapter matrices	shared	–	18.7/18.6†	45.7/43.9†	41.1±0.3^{†,p}
Continual NER	CPFD (2023)	previous model	shared	–	2.7/0.0†	0.3/15.1†	6.1±7.3†
	IS3 (2024)	previous model, class prototypes	shared	–	5.6/7.7†	0.0/40.6†	38.3†
Merging	LCA (2026)	base and per-session backbones, class Gaussians	shared	–	13.7/7.0†	38.0/36.4†	43.9±2.5†
This paper	CoLaR	50 docs/session, rank-128 layer-4 latents (100.7 MB)	shared	30.7±1.5	33.5±1.0	75.6/74.6†	87.8±0.5†
	CoLaR-SBR	50 docs/session, rank-128 layer-4 latents (100.7 MB)	blocked	89.1±2.0	88.4±2.6	**89.1±0.6**	86.2±1.3†
	+ LexSlot head slots	+ head slots, token ids	blocked	–	89.7±0.5†	89.6±1.0†	87.0±0.4†
	SBR-Recall	one rank-64 trunk delta per session (56.6 MB)	blocked, per delta	**92.6±1.0**	**91.9±0.6**	85.9^{†,b}	–

documents with 8-bit pixels, and 1.6× smaller only against raw documents as our code holds them. What it gives instead is a whole-document memory without pixels or token ids; latent replay needs the frozen lower layers, so it trains 45% of the parameters our raw run trains (a raw memory could freeze the same layers), with full-label F1 within 1.7 points of raw replay on every seed (Sec. 5.2). The rank is the one knob that sets its size; Appendix B gives the sizes at lower ranks, computed, not evaluated. SBR-Recall's cost grows with the sessions, not their documents: it is cheaper only when a session's documents take more than its delta.

7. Evidence beyond one stream

The comparison set. On CIL-CORD every method of Tab. 3 with a shared softmax stays at or below CoLaR in AA (33.5). Four baselines end below naive training on every seed because they learn later sessions poorly (Appendix C). On Mixed, where both causes act, CoLaR-SBR reaches 89.1±0.6 AA under the clean protocol; joint training and ER have only legacy cells there (87.7† and 74.5†, two seeds each), so we do not rank the three. Beyond CIL-CORD the evidence is the blocked readout with a memory on Mixed (three clean seeds), drift and replay on DIL (four encoders, legacy) and SBR-Recall's failure across schemas (one seed); the controls of H1 to H3 and the memory comparison ran on CIL-CORD only.

8. Limitations and conclusion

Limitations. Our study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones, and the ladder's controls ran on CIL-CORD only (Sec. 7). SBR-Recall on Mixed, the design-choice ablations and the image probe are single- or few-seed diagnostics. We did not run several comparators from this literature (among them PLOP, BPF, MiB's distillation and per-class sigmoid heads; Sec. 4.3). Joint training is a reference, not a bound, and ICE-O ran under our recipe, still improving at the epoch cap. SBR-Recall adds a delta and a pass per session and freezes its base after session 1, so it may depend on session order. CoLaR's memory grows with the sessions and at rank 128 takes more bytes than raw documents with 8-bit pixels; storing no raw pixels or token ids is weaker than privacy.

Conclusion. In this paper, we investigated why replay fails in continual KIE when the label space grows. The experiments show that two things go wrong and a memory repairs only one, drift; the other, the background conflict, lies almost entirely in stale stored labels: an oracle that keeps them true brings a shared softmax to within 2.4 points of joint training on every seed. Without true labels, a MiB-style loss recovers about half of the full-label gap, and a per-session readout most of it, by letting an "O" label count only against the classes its annotator could see; where label growth supplies a routing key, per-session deltas then replace the memory. We recommend testing continual document methods on a growing label space with a mask-free metric; in future work, we will run the ladder's controls on more streams and backbones.

LLM Usage Statement

LLM-based assistants were used in this work for writing and editing code (training scripts, analysis and figure-generation

scripts), for running and monitoring experiments, for searching and summarising related literature, and for drafting and revising the text of this paper, including an internal review of claims against the result files. All experiments were designed and decided by the authors, every reported number is produced by scripts from saved run files, and the authors reviewed and take full responsibility for all content.

References

[1] Hongjoon Ahn, Jihwan Kwak, Subin Lim, Hyeonsu Bang, Hyojun Kim, and Taesup Moon. SS-IL: Separated softmax for incremental learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 844–853, 2021. 2, 5

[2] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: A strong, simple baseline. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 2

[3] Lucas Caccia, Rahaf Aljundi, Nader Asadi, Tinne Tuytelaars, Joelle Pineau, and Eugene Belilovsky. New insights on reducing abrupt representation change in online continual learning. In *International Conference on Learning Representations (ICLR)*, 2022. 2

[4] Fabio Cermelli, Massimiliano Mancini, Samuel Rota Bulò, Elisa Ricci, and Barbara Caputo. Modeling the background for incremental learning in semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 2, 5

[5] Sungmin Cha and Kyunghyun Cho. Hyperparameters in continual learning: A reality check. *Transactions on Machine Learning Research (TMLR)*, 2025. 4

[6] Sungmin Cha, Beomyoung Kim, YoungJoon Yoo, and Taesup Moon. SSUL: Semantic segmentation with unknown label for exemplar-based class-incremental learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 2

[7] Arthur Douillard, Yifu Chen, Arnaud Dapogny, and Matthieu Cord. PLOP: Learning without forgetting for continual semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4040–4050, 2021. 2

[8] Nathan Greenberg, Trapit Bansal, Patrick Verga, and Andrew McCallum. Marginal likelihood training of BiLSTM-CRF for biomedical named entity recognition from disjoint label sets. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2824–2829, 2018. 5, 6

[9] Tyler L. Hayes, Kushal Kafle, Robik Shrestha, Manoj Acharya, and Christopher Kanan. REMIND your neural network to prevent catastrophic forgetting. In *European Conference on Computer Vision (ECCV)*, 2020. 2

[10] Jiangpeng He, Zhihao Duan, and Fengqing Zhu. CL-LoRA: Continual low-rank adaptation for rehearsal-free class-incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 30534–30544, 2025. 2

[11] Teakgyu Hong, Donghyun Kim, Mingi Ji, Wonseok Hwang, Daehyun Nam, and Sungrae Park. BROS: A pre-trained language model focusing on text and layout for better key information extraction from documents. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 10767–10775, 2022. 1, 2

[12] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. 7

[13] Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, and Furu Wei. LayoutLMv3: Pre-training for document AI with unified text and image masking. In *Proceedings of the 30th ACM International Conference on Multimedia (MM)*, pages 4083–4091, 2022. 1, 2

[14] Gyuhak Kim, Bing Liu, and Zixuan Ke. A multi-head model for continual learning via out-of-distribution replay. In *Proceedings of the First Conference on Lifelong Learning Agents (CoLLAs)*, pages 548–563, 2022. 2, 6

[15] Gyuhak Kim, Changnan Xiao, Tatsuya Konishi, Zixuan Ke, and Bing Liu. A theoretical study on solving continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 2, 7

[16] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences (PNAS)*, 114(13):3521–3526, 2017. 12

[17] Ritesh Kumar, Saurabh Goyal, Ashish Verma, and Vatche Isahagian. ProtoNER: Few shot incremental learning for named entity recognition using prototypical networks. In *Business Process Management Workshops (BPM 2023 International Workshops)*, pages 70–82. Springer, 2024. 2

[18] Siqi Li, Yufan Shen, Xiangnan Chen, Jiayi Chen, Hengwei Ju, Haodong Duan, Song Mao, Hongbin Zhou, Bo Zhang, Bin Fu, Pinlong Cai, Licheng Wen, Botian Shi, Yong Liu, Xinyu Cai, and Yu Qiao. GDI-Bench: A benchmark for general document intelligence with vision and reasoning decoupling. *arXiv preprint arXiv:2505.00063*, 2025. 2

[19] Zhizhong Li and Derek Hoiem. Learning without forgetting. In *European Conference on Computer Vision (ECCV)*, 2016. 2

[20] Minqian Liu and Lifu Huang. Teamwork is not always good: An empirical study of classifier drift in class-incremental information extraction. In *Findings of the Association for Computational Linguistics: ACL*, 2023. 2, 5, 6

[21] Yuyang Liu, Yang Cong, Dipam Goswami, Xialei Liu, and Joost van de Weijer. Augmented box replay: Overcoming foreground shift for incremental object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 2, 4

[22] Ruotian Ma, Xuanting Chen, Lin Zhang, Xin Zhou, Junzhe Wang, Tao Gui, Qi Zhang, Xiang Gao, and Yunwen Chen. Learning "O" helps for learning more: Handling the unlabeled entity problem for class-incremental NER. In *Proceedings of*

the 61st Annual Meeting of the Association for Computational Linguistics (ACL), pages 5959–5979, 2023. 2, 3

[23] Marc Masana, Xialei Liu, Bartłomiej Twardowski, Mikel Menta, Andrew D. Bagdanov, and Joost van de Weijer. Class-incremental learning: Survey and performance evaluation on image classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(5):5513–5533, 2023. 1

[24] Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165, 1989. 1

[25] Mark D. McDonnell, Dong Gong, Amin Parvaneh, Ehsan Abbasnejad, and Anton van den Hengel. RanPAC: Random projections and pre-trained models for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 15

[26] Mohammad Minouei, Khurram Azeem Hashmi, Mohammad Reza Soheili, Muhammad Zeshan Afzal, and Didier Stricker. Continual learning for table detection in document images. *Applied Sciences*, 12(18):8969, 2022. 2

[27] Qijie Mo, Yipeng Gao, Shenghao Fu, Junkai Yan, Ancong Wu, and Wei-Shi Zheng. Bridge past and future: Overcoming information asymmetry in incremental object detection. In *European Conference on Computer Vision (ECCV)*, pages 463–480. Springer, 2024. 2, 5

[28] Natawut Monaikul, Giuseppe Castellucci, Simone Filice, and Oleg Rokhlenko. Continual learning for named entity recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 13570–13577, 2021. 2

[29] Lorenzo Pellegrini, Gabriele Graffieti, Vincenzo Lomonaco, and Davide Maltoni. Latent replay for real-time continual learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020. 2

[30] Shengjie Qiu, Junhao Zheng, Zhen Liu, Yicheng Luo, and Qianli Ma. Incremental sequence labeling: A tale of two shifts. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 777–791, 2024. arXiv:2402.10447. 3

[31] Vinay V. Ramasesh, Ethan Dyer, and Maithra Raghu. Anatomy of catastrophic forgetting: Hidden representations and task semantics. In *International Conference on Learning Representations (ICLR)*, 2021. 14

[32] David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy P. Lillicrap, and Greg Wayne. Experience replay for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 2

[33] Abdellatif Sassioui, Yasser Elouargui, Mohamed El Kamili, Rachid Benouini, El Mehdi Benyoussef, Meriyem Chergui, and Mohammed Ouzzif. AMD-Proj: Adaptive memory-driven selective gradient projection for continual learning in document understanding. *Technologies*, 14(5):250, 2026. 2

[34] Abdellatif Sassioui, Yasser Elouargui, Mohamed El Kamili, Rachid Benouini, El Mehdi Benyoussef, Meriyem Chergui, and Mohamed Ouzzif. When documents evolve: Decomposing catastrophic forgetting in layout-aware document models. In *International Conference on Wireless Communications and Mobile Computing (IWCMC)*, pages 1124–1129, 2026. 2

[35] James Seale Smith, Leonid Karlinsky, Vyshnavi Gutta, Paola Cascante-Bonilla, Donghyun Kim, Assaf Arbelle, Rameswar Panda, Rogerio Feris, and Zsolt Kira. CODA-Prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[36] Tung Tran, Danilo Vasconcellos Vargas, and Khoat Than. LCA: Local classifier alignment for continual learning. In *International Conference on Learning Representations (ICLR)*, 2026. 2

[37] Jiapeng Wang, Lianwen Jin, and Kai Ding. LiLT: A simple yet effective language-independent layout transformer for structured document understanding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. 1, 2

[38] Liyuan Wang, Xingxing Zhang, Hang Su, and Jun Zhu. A comprehensive survey of continual learning: Theory, method and application. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 46(8):5362–5383, 2024. 1, 2

[39] Xiao Wang, Tianze Chen, Qiming Ge, Han Xia, Rong Bao, Rui Zheng, Qi Zhang, Tao Gui, and Xuanjing Huang. Orthogonal subspace learning for language model continual learning. In *Findings of the Association for Computational Linguistics: EMNLP*, 2023. 12

[40] Zifeng Wang, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. DualPrompt: Complementary prompting for rehearsal-free continual learning. In *European Conference on Computer Vision (ECCV)*, 2022. 12

[41] Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. Learning to prompt for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 12

[42] Yuexiang Zhai, Shengbang Tong, Xiao Li, Mu Cai, Qing Qu, Yong Jae Lee, and Yi Ma. Investigating the catastrophic forgetting in multimodal large language model fine-tuning. In *Proceedings of the Conference on Parsimony and Learning (CPAL)*, pages 202–227, 2024. 2

[43] Duzhen Zhang, Wei Cong, Jiahua Dong, Yahan Yu, Xiuyi Chen, Yonggang Zhang, and Zhen Fang. Continual named entity recognition without catastrophic forgetting. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. 2, 3

[44] Gengwei Zhang, Liyuan Wang, Guoliang Kang, Ling Chen, and Yunchao Wei. SLCA: Slow learner with classifier alignment for continual learning on a pre-trained model. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 19091–19101, 2023. 15